



Quality Control Report

Report ID: QC-S73932-01
Generated on 05/16/2025

Report to

Attention: Cindy Tomaszewski
City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368 FAX:

Report Produced by

Merit Laboratories
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Report Summary

Lab Sample ID(s): S73932.01
Project: Quarterly Biosolids
Submitted Date/Time: 04/29/2025 10:00
Sampled by: Unknown
P.O. #:

QC Report Sections

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Report Flag Descriptions

*: QC result is outside of indicated control limits
W: Surrogate result not applicable due to sample dilution

I certify that this data package is in compliance with the terms and conditions of the program, and project, and contractual requirements both technically and for completeness. Release of the data contained in this hardcopy data package and its computer-readable data submitted has been authorized by the Quality Assurance Manager and his/her designee, as verified by the following signature.

Barbara Ball
Quality Assurance Manager

QC Report - Analysis Summary

Lab Sample ID: S73932.01

Sample Tag: Digester #4

Collected Date/Time: 04/28/2025 12:54

Matrix: Sludge

COC Reference: 175249

Analysis	Method	Run Date/Time	Batch ID	Prep ID	Surr	QC Types
<i>Inorganics</i>						
Total Solids	SM2540B	04/30/25 14:15	TS250430A	TS250430A	No	BLK/LCS/DUP
<i>Organics - Volatiles</i>						
PFAs EPA Method 1633	EPA 1633A	05/13/25 14:17	CI250513_1633	E1633250512S1	Yes	BLK/LCS/LCSD
PFAs EPA Method 1633 (Replicate 01)	EPA 1633A	05/08/25 16:10	CI250508_1633	E1633250507S1	Yes	BLK/LCS/LCSD

QC Report - Prep Batch Summary

Inorganics, Prep Batch ID: TS250430A

Surrogates: No, QC Types: BLK/LCS/DUP

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S73932.01	Total Solids	SM2540B	04/30/25 14:15	TS250430A

Organics - Volatiles, Prep Batch ID: E1633250507S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S73932.01	PFA's EPA Method 1633 (Replicate 01)	EPA 1633A	05/08/25 16:10	CI250508_1633

Organics - Volatiles, Prep Batch ID: E1633250512S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Sample ID	Analysis	Method	Run Date/Time	Batch ID
S73932.01	PFA's EPA Method 1633	EPA 1633A	05/13/25 14:17	CI250513_1633

QC Report - Surrogates per Lab Sample

Lab Sample ID: S73932.01

Sample Tag: Digester #4

Collected Date/Time: 04/28/2025 12:54

Matrix: Sludge

COC Reference: 175249

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250513_1633, Run Date: 05/13/2025 14:17, Matrix: SO, Dilution: 19.1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		92.8	8	130
M5PFPeA-S		89.6	35	130
M2-4:2FTSA-S	*	232.4	40	165
M5PFHxA-S		90.3	40	130
M3PFBS-S		91.3	40	135
MHFPO-DA-S		78.7	40	130
M4PFHpA-S		89.7	40	130
M2-6:2FTSA-S	*	252.5	40	215
M8PFOA-S		88.4	40	130
M3PFHxS-S		89.7	40	130
M9-PFNA-S		91.0	40	130
M2-8:2FTSA-S		225.3	40	275
d3N-MeFOSAA-S		90.5	40	135
M6PFDA-S		87.2	40	130
d5EtFOSAA-S		81.2	40	150
M8PFOS-S		82.1	40	130
M7PFUnDA-S		50.9	40	130
MPFDoDA-S	*	22.0	40	130
M8FOSA-S		84.2	40	130
M2PFTeDA-S	*	5.8	20	130
d7-N-MeFOSE-M-S		48.5	20	130
d-N-MeFOSA-M-S		34.1	10	130
d9-N-EtFOSE-M-S		40.9	15	130
d-N-EtFOSA-M-S		19.7	10	130

Organics - Volatiles, Analysis: PFAs EPA Method 1633 (Replicate 01)

Run in Batch: CI250508_1633, Run Date: 05/08/2025 16:10, Matrix: SO, Dilution: 129

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.5	8	130
M5PFPeA-S		92.5	35	130
M2-4:2FTSA-S		138.0	40	165
M5PFHxA-S		91.2	40	130
M3PFBS-S		92.5	40	135
MHFPO-DA-S		91.3	40	130
M4PFHpA-S		90.4	40	130
M2-6:2FTSA-S		150.6	40	215
M8PFOA-S		90.1	40	130
M3PFHxS-S		88.3	40	130
M9-PFNA-S		95.5	40	130
M2-8:2FTSA-S		160.9	40	275
d3N-MeFOSAA-S		107.3	40	135
M6PFDA-S		93.2	40	130
d5EtFOSAA-S		115.9	40	150
M8PFOS-S		79.2	40	130
M7PFUnDA-S		84.3	40	130
MPFDoDA-S		70.1	40	130
M8FOSA-S		84.7	40	130

QC Report - Surrogates per Lab Sample

M2PFTeDA-S	31.2	20	130
d7-N-MeFOSE-M-S	80.0	20	130
d-N-MeFOSA-M-S	74.4	10	130
d9-N-EtFOSE-M-S	79.7	15	130
d-N-EtFOSA-M-S	53.6	10	130

QC Report - Surrogates per QC Sample

Organics - Volatiles, Prep Batch ID: E1633250507S1

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: CI250508_1633.BLK250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 14:24, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		94.0	8	130
M5PFPeA-S		93.3	35	130
M2-4:2FTSA-S		113.4	40	165
M5PFHxA-S		91.2	40	130
M3PFBS-S		94.7	40	135
MHFPO-DA-S		91.4	40	130
M4PFHpA-S		90.9	40	130
M2-6:2FTSA-S		108.2	40	215
M8PFOA-S		85.7	40	130
M3PFHxS-S		91.1	40	130
M9-PFNA-S		94.0	40	130
M2-8:2FTSA-S		109.9	40	275
d3N-MeFOSAA-S		83.1	40	135
M6PFDA-S		87.2	40	130
d5EtFOSAA-S		84.5	40	150
M8PFOS-S		87.3	40	130
M7PFUnDA-S		79.4	40	130
MPFDoDA-S		80.7	40	130
M8FOSA-S		78.6	40	130
M2PFTeDA-S		85.0	20	130
d7-N-MeFOSE-M-S		68.9	20	130
d-N-MeFOSA-M-S		66.1	10	130
d9-N-EtFOSE-M-S		66.3	15	130
d-N-EtFOSA-M-S		60.7	10	130

Laboratory Control Sample (LCS)

Lab Sample ID: CI250508_1633.LCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 15:06, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		85.2	8	130
M5PFPeA-S		91.4	35	130
M2-4:2FTSA-S		103.1	40	165
M5PFHxA-S		88.5	40	130
M3PFBS-S		90.4	40	135
MHFPO-DA-S		89.1	40	130
M4PFHpA-S		88.0	40	130
M2-6:2FTSA-S		101.7	40	215
M8PFOA-S		84.5	40	130
M3PFHxS-S		90.2	40	130
M9-PFNA-S		86.4	40	130
M2-8:2FTSA-S		106.1	40	275
d3N-MeFOSAA-S		90.0	40	135
M6PFDA-S		91.3	40	130
d5EtFOSAA-S		93.9	40	150
M8PFOS-S		85.3	40	130
M7PFUnDA-S		85.4	40	130
MPFDoDA-S		85.4	40	130
M8FOSA-S		83.3	40	130
M2PFTeDA-S		83.3	20	130

QC Report - Surrogates per QC Sample

d7-N-MeFOSE-M-S	63.4	20	130
d-N-MeFOSA-M-S	66.8	10	130
d9-N-EtFOSE-M-S	60.8	15	130
d-N-EtFOSA-M-S	62.9	10	130

QC Report - Surrogates per QC Sample

Laboratory Control Sample (LCS)

Lab Sample ID: CI250508_1633.LLLCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 14:45, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		90.9	8	130
M5PFPeA-S		90.3	35	130
M2-4:2FTSA-S		103.6	40	165
M5PFHxA-S		88.4	40	130
M3PFBS-S		91.4	40	135
MHFPO-DA-S		88.8	40	130
M4PFHpA-S		89.8	40	130
M2-6:2FTSA-S		101.1	40	215
M8PFOA-S		85.6	40	130
M3PFHxS-S		85.8	40	130
M9-PFNA-S		90.3	40	130
M2-8:2FTSA-S		110.2	40	275
d3N-MeFOSAA-S		81.4	40	135
M6PFDA-S		84.7	40	130
d5EtFOSAA-S		80.7	40	150
M8PFOS-S		82.5	40	130
M7PFUnDA-S		78.5	40	130
MPFDoDA-S		76.7	40	130
M8FOSA-S		79.7	40	130
M2PFTeDA-S		82.6	20	130
d7-N-MeFOSE-M-S		66.5	20	130
d-N-MeFOSA-M-S		61.4	10	130
d9-N-EtFOSE-M-S		63.8	15	130
d-N-EtFOSA-M-S		62.1	10	130

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: CI250508_1633.LCSD250507, Parent Sample ID: CI250508_1633.LCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 15:27, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		90.8	8	130
M5PFPeA-S		89.8	35	130
M2-4:2FTSA-S		104.9	40	165
M5PFHxA-S		89.0	40	130
M3PFBS-S		89.0	40	135
MHFPO-DA-S		88.0	40	130
M4PFHpA-S		87.1	40	130
M2-6:2FTSA-S		99.7	40	215
M8PFOA-S		84.8	40	130
M3PFHxS-S		89.5	40	130
M9-PFNA-S		87.8	40	130
M2-8:2FTSA-S		97.5	40	275
d3N-MeFOSAA-S		79.5	40	135
M6PFDA-S		85.6	40	130
d5EtFOSAA-S		83.8	40	150
M8PFOS-S		84.7	40	130
M7PFUnDA-S		77.8	40	130
MPFDoDA-S		78.7	40	130
M8FOSA-S		82.1	40	130
M2PFTeDA-S		86.0	20	130
d7-N-MeFOSE-M-S		66.8	20	130
d-N-MeFOSA-M-S		62.8	10	130
d9-N-EtFOSE-M-S		63.6	15	130

QC Report - Surrogates per QC Sample

d-N-EtFOSA-M-S

59.9

10

130

QC Report - Surrogates per QC Sample

Organics - Volatiles, Prep Batch ID: E1633250512S1

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: CI250513_1633.BLK250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 12:52, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		99.1	8	130
M5PFPeA-S		101.4	35	130
M2-4:2FTSA-S		107.5	40	165
M5PFHxA-S		96.4	40	130
M3PFBS-S		98.7	40	135
MHFPO-DA-S		96.0	40	130
M4PFHpA-S		97.5	40	130
M2-6:2FTSA-S		103.3	40	215
M8PFOA-S		94.5	40	130
M3PFHxS-S		98.7	40	130
M9-PFNA-S		99.8	40	130
M2-8:2FTSA-S		104.9	40	275
d3N-MeFOSAA-S		91.9	40	135
M6PFDA-S		95.4	40	130
d5EtFOSAA-S		88.4	40	150
M8PFOS-S		91.9	40	130
M7PFUnDA-S		90.9	40	130
MPFDoDA-S		89.1	40	130
M8FOSA-S		90.6	40	130
M2PFTeDA-S		94.5	20	130
d7-N-MeFOSE-M-S		82.0	20	130
d-N-MeFOSA-M-S		75.6	10	130
d9-N-EtFOSE-M-S		81.2	15	130
d-N-EtFOSA-M-S		76.0	10	130

Laboratory Control Sample (LCS)

Lab Sample ID: CI250513_1633.LCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:35, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		93.6	8	130
M5PFPeA-S		96.9	35	130
M2-4:2FTSA-S		96.4	40	165
M5PFHxA-S		90.8	40	130
M3PFBS-S		90.9	40	135
MHFPO-DA-S		89.6	40	130
M4PFHpA-S		91.5	40	130
M2-6:2FTSA-S		92.4	40	215
M8PFOA-S		88.6	40	130
M3PFHxS-S		93.8	40	130
M9-PFNA-S		92.5	40	130
M2-8:2FTSA-S		95.2	40	275
d3N-MeFOSAA-S		86.6	40	135
M6PFDA-S		90.0	40	130
d5EtFOSAA-S		87.3	40	150
M8PFOS-S		85.6	40	130
M7PFUnDA-S		84.0	40	130
MPFDoDA-S		79.3	40	130
M8FOSA-S		90.4	40	130
M2PFTeDA-S		83.6	20	130

QC Report - Surrogates per QC Sample

d7-N-MeFOSE-M-S	66.9	20	130
d-N-MeFOSA-M-S	62.8	10	130
d9-N-EtFOSE-M-S	61.5	15	130
d-N-EtFOSA-M-S	56.0	10	130

QC Report - Surrogates per QC Sample

Laboratory Control Sample (LCS)

Lab Sample ID: CI250513_1633.LLLCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:14, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		93.4	8	130
M5PFPeA-S		94.6	35	130
M2-4:2FTSA-S		96.0	40	165
M5PFHxA-S		90.7	40	130
M3PFBS-S		89.5	40	135
MHFPO-DA-S		89.1	40	130
M4PFHpA-S		89.4	40	130
M2-6:2FTSA-S		94.2	40	215
M8PFOA-S		90.1	40	130
M3PFHxS-S		88.8	40	130
M9-PFNA-S		93.7	40	130
M2-8:2FTSA-S		99.7	40	275
d3N-MeFOSAA-S		88.3	40	135
M6PFDA-S		92.0	40	130
d5EtFOSAA-S		91.4	40	150
M8PFOS-S		90.1	40	130
M7PFUnDA-S		85.6	40	130
MPFDoDA-S		84.5	40	130
M8FOSA-S		89.9	40	130
M2PFTeDA-S		77.5	20	130
d7-N-MeFOSE-M-S		75.1	20	130
d-N-MeFOSA-M-S		70.8	10	130
d9-N-EtFOSE-M-S		71.6	15	130
d-N-EtFOSA-M-S		67.7	10	130

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: CI250513_1633.LCSD250512, Parent Sample ID: CI250513_1633.LCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:56, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Surrogate	Flags	%Rec	LCL	UCL
MPFBA-S		95.4	8	130
M5PFPeA-S		94.1	35	130
M2-4:2FTSA-S		98.6	40	165
M5PFHxA-S		91.4	40	130
M3PFBS-S		93.0	40	135
MHFPO-DA-S		92.5	40	130
M4PFHpA-S		92.2	40	130
M2-6:2FTSA-S		99.6	40	215
M8PFOA-S		90.4	40	130
M3PFHxS-S		93.4	40	130
M9-PFNA-S		94.4	40	130
M2-8:2FTSA-S		98.3	40	275
d3N-MeFOSAA-S		87.0	40	135
M6PFDA-S		89.5	40	130
d5EtFOSAA-S		85.6	40	150
M8PFOS-S		88.7	40	130
M7PFUnDA-S		86.4	40	130
MPFDoDA-S		84.3	40	130
M8FOSA-S		89.5	40	130
M2PFTeDA-S		83.1	20	130
d7-N-MeFOSE-M-S		74.1	20	130
d-N-MeFOSA-M-S		69.5	10	130
d9-N-EtFOSE-M-S		71.5	15	130

QC Report - Surrogates per QC Sample

d-N-EtFOSA-M-S

64.5

10

130

QC Report - Internal Standards per Lab Sample

Lab Sample ID: S73932.01

Sample Tag: Digester #4

Collected Date/Time: 04/28/2025 12:54

Matrix: Sludge

COC Reference: 175249

Organics - Volatiles, Analysis: PFAs EPA Method 1633

Run in Batch: CI250513_1633, Run Date: 05/13/2025 14:17, Matrix: SO, Dilution: 19.1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		113.7	50	200.0
M2PFHxA		111.5	50	200.0
M4PFOA		114.6	50	200.0
M5-PFNA		112.2	50	200.0
M4PFOS		88.2	50	200.0
M2PFDA		101.1	50	200.0
O2PFHxS		98.5	50	200.0

Organics - Volatiles, Analysis: PFAs EPA Method 1633 (Replicate 01)

Run in Batch: CI250508_1633, Run Date: 05/08/2025 16:10, Matrix: SO, Dilution: 129

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		102.2	50	200.0
M2PFHxA		98.2	50	200.0
M4PFOA		100.5	50	200.0
M5-PFNA		97.5	50	200.0
M4PFOS		109.3	50	200.0
M2PFDA		96.8	50	200.0
O2PFHxS		97.8	50	200.0

QC Report - Internal Standards per QC Sample

Organics - Volatiles, Prep Batch ID: E1633250507S1

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: CI250508_1633.BLK250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 14:24, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		96.4	50	200.0
M2PFHxA		94.7	50	200.0
M4PFOA		103.1	50	200.0
M5-PFNA		96.1	50	200.0
M4PFOS		100.3	50	200.0
M2PFDA		97.2	50	200.0
O2PFHxS		95.1	50	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250508_1633.LCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 15:06, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		101.2	50	200.0
M2PFHxA		98.5	50	200.0
M4PFOA		106.7	50	200.0
M5-PFNA		104.1	50	200.0
M4PFOS		106.3	50	200.0
M2PFDA		100.1	50	200.0
O2PFHxS		100.8	50	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250508_1633.LLLCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 14:45, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		91.8	50	200.0
M2PFHxA		89.8	50	200.0
M4PFOA		95.7	50	200.0
M5-PFNA		93.2	50	200.0
M4PFOS		97.7	50	200.0
M2PFDA		96.5	50	200.0
O2PFHxS		94.3	50	200.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: CI250508_1633.LCSD250507, Parent Sample ID: CI250508_1633.LCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 15:27, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		95.0	50	200.0
M2PFHxA		94.1	50	200.0
M4PFOA		99.2	50	200.0
M5-PFNA		96.0	50	200.0
M4PFOS		99.1	50	200.0
M2PFDA		97.9	50	200.0
O2PFHxS		97.3	50	200.0

QC Report - Internal Standards per QC Sample

Organics - Volatiles, Prep Batch ID: E1633250512S1

QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: CI250513_1633.BLK250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 12:52, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		103.3	50	200.0
M2PFHxA		95.1	50	200.0
M4PFOA		101.2	50	200.0
M5-PFNA		96.2	50	200.0
M4PFOS		101.1	50	200.0
M2PFDA		98.6	50	200.0
O2PFHxS		93.7	50	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250513_1633.LCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:35, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		106.2	50	200.0
M2PFHxA		98.3	50	200.0
M4PFOA		106.0	50	200.0
M5-PFNA		103.8	50	200.0
M4PFOS		106.9	50	200.0
M2PFDA		104.9	50	200.0
O2PFHxS		98.4	50	200.0

Laboratory Control Sample (LCS)

Lab Sample ID: CI250513_1633.LLLCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:14, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		106.4	50	200.0
M2PFHxA		99.5	50	200.0
M4PFOA		104.8	50	200.0
M5-PFNA		101.5	50	200.0
M4PFOS		104.0	50	200.0
M2PFDA		102.2	50	200.0
O2PFHxS		101.9	50	200.0

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: CI250513_1633.LCSD250512, Parent Sample ID: CI250513_1633.LCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:56, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Internal Standard	Flags	%Rec	LCL	UCL
M3PFBA		108.4	50	200.0
M2PFHxA		101.0	50	200.0
M4PFOA		108.7	50	200.0
M5-PFNA		104.7	50	200.0
M4PFOS		108.0	50	200.0
M2PFDA		107.2	50	200.0
O2PFHxS		100.9	50	200.0

QC Report - Batch QC Results

Inorganics, Prep Batch ID: TS250430A

Surrogates: No, QC Types: BLK/LCS/DUP

Blank (BLK)

Lab Sample ID: TS250430A.LRB1

Run in Batch: TS250430A, Run Date: 04/30/2025 14:15, Prep Date: 04/30/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	Conc	RDL	Units
Total Solids		ND	1	%

Laboratory Control Sample (LCS)

Lab Sample ID: TS250430A.LCS1

Run in Batch: TS250430A, Run Date: 04/30/2025 14:15, Prep Date: 04/30/2025, Matrix: Liquid, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
Total Solids		100	90	110

Duplicate (DUP)

Lab Sample ID: TS250430A.DP1, Parent Sample ID: S73663.01

Run in Batch: TS250430A, Run Date: 04/30/2025 14:15, Prep Date: 04/30/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		2	5

Duplicate (DUP)

Lab Sample ID: TS250430A.DP2, Parent Sample ID: S73864.02

Run in Batch: TS250430A, Run Date: 04/30/2025 14:15, Prep Date: 04/30/2025, Matrix: Soil, Dilution: 1

Analyte	Flags	RPD	RPD CL
Total Solids		3	5

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250507S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: CI250508_1633.BLK250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 14:24, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	100	ng/kg
PFMPA		ND	100	ng/kg
FPrPA (3:3 FTCA)		ND	200	ng/kg
PFPeA		ND	100	ng/kg
PFMBA		ND	100	ng/kg
4:2 FTSA		ND	100	ng/kg
NFDHA		ND	100	ng/kg
PFHxA		ND	100	ng/kg
PFBS		ND	100	ng/kg
HFPO-DA		ND	100	ng/kg
FPePA (5:3 FTCA)		ND	200	ng/kg
PFEESA		ND	100	ng/kg
PFHpA		ND	100	ng/kg
ADONA		ND	100	ng/kg
PFPeS		ND	100	ng/kg
6:2 FTSA		ND	100	ng/kg
PFOA		ND	100	ng/kg
PFHxS		ND	100	ng/kg
FHpPA (7:3 FTCA)		ND	200	ng/kg
PFNA		ND	100	ng/kg
8:2 FTSA		ND	100	ng/kg
PFHpS		ND	100	ng/kg
N-MeFOSAA		ND	100	ng/kg
PFDA		ND	100	ng/kg
EtFOSAA		ND	100	ng/kg
PFOS		ND	100	ng/kg
PFUnDA		ND	100	ng/kg
9CL-PF3ONS		ND	100	ng/kg
PFNS		ND	100	ng/kg
PFDODA		ND	100	ng/kg
PFDS		ND	100	ng/kg
PFTTrDA		ND	100	ng/kg
11CL-PF3OUdS		ND	100	ng/kg
FOSA		ND	100	ng/kg
PFTeDA		ND	100	ng/kg
PFDOS		ND	100	ng/kg
NMeFOSE		ND	100	ng/kg
NMeFOSAM		ND	100	ng/kg
NEtFOSE		ND	100	ng/kg
NEtFOSAM		ND	100	ng/kg

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250507S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250508_1633.LCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 15:06, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		107.2	70	140
PFMPA		94.5	30	140
FPrPA (3:3 FTCA)		82.1	45	130
PFPeA		109.6	60	150
PFMBA		105.8	60	150
4:2 FTSA		109.7	60	150
NFDHA		107.3	60	155
PFHxA		107.0	65	140
PFBS		109.7	65	145
HFPO-DA		103.9	70	145
FPePA (5:3 FTCA)		86.8	60	130
PFEESA		113.9	70	140
PFHpA		111.4	65	145
ADONA		110.2	70	160
PFPeS		107.6	55	160
6:2 FTSA		106.8	55	200
PFOA		103.5	70	150
PFHxS		106.3	60	150
FHpPA (7:3 FTCA)		95.2	60	150
PFNA		97.4	70	155
8:2 FTSA		100.0	70	150
PFHpS		108.5	65	155
N-MeFOSAA		103.0	65	155
PFDA		109.9	70	155
EtFOSAA		102.2	65	165
PFOS		103.7	65	160
PFUnDA		106.6	70	155
9CL-PF3ONS		111.7	70	150
PFNS		105.4	55	140
PFDODA		99.9	70	150
PFDS		102.5	40	155
PFTTrDA		94.7	65	150
11CL-PF3OUdS		101.5	45	160
FOSA		106.6	70	140
PFTeDA		107.5	65	150
PFDOS		94.3	25	160
NMeFOSE		103.4	70	140
NMeFOSAM		101.4	70	155
NEtFOSE		105.4	65	165
NEtFOSAM		101.3	70	140

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250507S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250508_1633.LLLCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 14:45, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		102.8	70	140
PFMPA		108.3	30	140
FPrPA (3:3 FTCA)		91.1	45	130
PFPeA		107.9	60	150
PFMBA		99.2	60	150
4:2 FTSA		101.1	60	150
NFDHA		108.1	60	155
PFHxA		103.5	65	140
PFBS		102.6	65	145
HFPO-DA		101.6	70	145
FPePA (5:3 FTCA)		103.4	60	130
PFEESA		108.7	70	140
PFHpA		103.9	65	145
ADONA		108.9	70	160
PFPeS		105.6	55	160
6:2 FTSA		102.5	55	200
PFOA		106.7	70	150
PFHxS		102.3	60	150
FHpPA (7:3 FTCA)		83.3	60	150
PFNA		108.9	70	155
8:2 FTSA		86.5	70	150
PFHpS		105.7	65	155
N-MeFOSAA		99.6	65	155
PFDA		103.0	70	155
EtFOSAA		105.5	65	165
PFOS		105.6	65	160
PFUnDA		100.0	70	155
9CL-PF3ONS		105.1	70	150
PFNS		101.2	55	140
PFDODA		104.1	70	150
PFDS		90.0	40	155
PFTTrDA		106.8	65	150
11CL-PF3OUdS		91.4	45	160
FOSA		105.8	70	140
PFTeDA		103.7	65	150
PFDOS		101.7	25	160
NMeFOSE		118.7	70	140
NMeFOSAM		115.6	70	155
NEtFOSE		103.2	65	165
NEtFOSAM		108.8	70	140

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250507S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: CI250508_1633.LCSD250507, Parent Sample ID: CI250508_1633.LCS250507

Run in Batch: CI250508_1633, Run Date: 05/08/2025 15:27, Prep Date: 05/07/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		105.8	70	140	1.3	30.0
PFMPA		96.5	30	140	2.1	30.0
FPrPA (3:3 FTCA)		87.6	45	130	6.4	30.0
PFPeA		110.4	60	150	0.7	30.0
PFMBA		105.5	60	150	0.3	30.0
4:2 FTSA		105.4	60	150	3.9	30.0
NFDHA		98.3	60	155	8.7	30.0
PFHxA		106.9	65	140	0.1	30.0
PFBS		111.6	65	145	1.7	30.0
HFPO-DA		103.5	70	145	0.3	30.0
FPePA (5:3 FTCA)		82.4	60	130	5.2	30.0
PFEESA		109.4	70	140	4.0	30.0
PFHpA		112.5	65	145	0.9	30.0
ADONA		110.7	70	160	0.5	30.0
PFPeS		104.1	55	160	3.3	30.0
6:2 FTSA		103.5	55	200	3.1	30.0
PFOA		100.6	70	150	2.8	30.0
PFHxS		101.5	60	150	4.6	30.0
FHpPA (7:3 FTCA)		85.0	60	150	11.3	30.0
PFNA		98.9	70	155	1.4	30.0
8:2 FTSA		106.4	70	150	6.2	30.0
PFHpS		109.5	65	155	0.9	30.0
N-MeFOSAA		107.1	65	155	3.9	30.0
PFDA		110.8	70	155	0.9	30.0
EtFOSAA		100.3	65	165	1.8	30.0
PFOS		104.4	65	160	0.7	30.0
PFUnDA		108.7	70	155	1.9	30.0
9CL-PF3ONS		102.4	70	150	8.7	30.0
PFNS		103.6	55	140	1.7	30.0
PFDODA		99.0	70	150	0.9	30.0
PFDS		95.9	40	155	6.7	30.0
PFTTrDA		103.9	65	150	9.3	30.0
11CL-PF3OUdS		95.0	45	160	6.5	30.0
FOSA		106.8	70	140	0.1	30.0
PFTeDA		110.2	65	150	2.5	30.0
PFDOS		103.7	25	160	9.5	30.0
NMeFOSE		106.8	70	140	3.2	30.0
NMeFOSAM		110.2	70	155	8.3	30.0
NEtFOSE		104.5	65	165	0.9	30.0
NEtFOSAM		99.8	70	140	1.6	30.0

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250512S1

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Blank (BLK)

Lab Sample ID: CI250513_1633.BLK250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 12:52, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Analyte	Flags	Conc	RDL	Units
PFBA		ND	100	ng/kg
PFMPA		ND	100	ng/kg
FPrPA (3:3 FTCA)		ND	200	ng/kg
PFPeA		ND	100	ng/kg
PFMBA		ND	100	ng/kg
4:2 FTSA		ND	100	ng/kg
NFDHA		ND	100	ng/kg
PFHxA		ND	100	ng/kg
PFBS		ND	100	ng/kg
HFPO-DA		ND	100	ng/kg
FPePA (5:3 FTCA)		ND	200	ng/kg
PFEESA		ND	100	ng/kg
PFHpA		ND	100	ng/kg
ADONA		ND	100	ng/kg
PFPeS		ND	100	ng/kg
6:2 FTSA		ND	100	ng/kg
PFOA		ND	100	ng/kg
PFHxS		ND	100	ng/kg
FHpPA (7:3 FTCA)		ND	200	ng/kg
PFNA		ND	100	ng/kg
8:2 FTSA		ND	100	ng/kg
PFHpS		ND	100	ng/kg
PFDA		ND	100	ng/kg
N-MeFOSAA		ND	100	ng/kg
EtFOSAA		ND	100	ng/kg
PFOS		ND	100	ng/kg
PFUnDA		ND	100	ng/kg
9CL-PF3ONS		ND	100	ng/kg
PFNS		ND	100	ng/kg
PFDODA		ND	100	ng/kg
PFDS		ND	100	ng/kg
PFTTrDA		ND	100	ng/kg
11CL-PF3OUdS		ND	100	ng/kg
FOSA		ND	100	ng/kg
PFTeDA		ND	100	ng/kg
PFDOS		ND	100	ng/kg
NMeFOSE		ND	100	ng/kg
NMeFOSAM		ND	100	ng/kg
NEtFOSE		ND	100	ng/kg
NEtFOSAM		ND	100	ng/kg

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250512S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250513_1633.LCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:35, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		107.7	70	140
PFMPA		100.3	30	140
FPrPA (3:3 FTCA)		98.8	45	130
PFPeA		111.7	60	150
PFMBA		105.8	60	150
4:2 FTSA		104.9	60	150
NFDHA		104.1	60	155
PFHxA		107.3	65	140
PFBS		107.5	65	145
HFPO-DA		105.0	70	145
FPePA (5:3 FTCA)		106.2	60	130
PFEESA		111.2	70	140
PFHpA		109.6	65	145
ADONA		114.2	70	160
PFPeS		101.6	55	160
6:2 FTSA		112.3	55	200
PFOA		101.3	70	150
PFHxS		106.4	60	150
FHpPA (7:3 FTCA)		115.1	60	150
PFNA		98.1	70	155
8:2 FTSA		97.9	70	150
PFHpS		104.7	65	155
PFDA		105.3	70	155
N-MeFOSAA		99.8	65	155
EtFOSAA		103.9	65	165
PFOS		106.4	65	160
PFUnDA		105.6	70	155
9CL-PF3ONS		113.3	70	150
PFNS		103.0	55	140
PFDODA		97.3	70	150
PFDS		96.3	40	155
PFTTrDA		97.3	65	150
11CL-PF3OUdS		97.2	45	160
FOSA		107.5	70	140
PFTeDA		104.8	65	150
PFDOS		98.2	25	160
NMeFOSE		104.8	70	140
NMeFOSAM		114.7	70	155
NEtFOSE		105.3	65	165
NEtFOSAM		101.4	70	140

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250512S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample (LCS)

Lab Sample ID: CI250513_1633.LLLCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:14, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL
PFBA		101.9	70	140
PFMPA		106.5	30	140
FPrPA (3:3 FTCA)		126.7	45	130
PFPeA		105.2	60	150
PFMBA		101.8	60	150
4:2 FTSA		95.5	60	150
NFDHA		81.6	60	155
PFHxA		106.7	65	140
PFBS		103.2	65	145
HFPO-DA		92.9	70	145
FPePA (5:3 FTCA)		119.4	60	130
PFEESA		99.9	70	140
PFHpA		108.1	65	145
ADONA		109.5	70	160
PFPeS		98.4	55	160
6:2 FTSA		99.5	55	200
PFOA		102.0	70	150
PFHxS		103.0	60	150
FHpPA (7:3 FTCA)		92.9	60	150
PFNA		101.2	70	155
8:2 FTSA		95.2	70	150
PFHpS		95.2	65	155
PFDA		102.4	70	155
N-MeFOSAA		96.2	65	155
EtFOSAA		94.5	65	165
PFOS		102.9	65	160
PFUnDA		101.4	70	155
9CL-PF3ONS		104.2	70	150
PFNS		98.9	55	140
PFDODA		102.6	70	150
PFDS		98.5	40	155
PFTTrDA		98.7	65	150
11CL-PF3OUdS		93.3	45	160
FOSA		100.7	70	140
PFTeDA		102.0	65	150
PFDOS		90.4	25	160
NMeFOSE		111.8	70	140
NMeFOSAM		109.3	70	155
NEtFOSE		103.7	65	165
NEtFOSAM		108.1	70	140

QC Report - Batch QC Results

Organics - Volatiles, Prep Batch ID: E1633250512S1 (continued)

Surrogates: Yes, QC Types: BLK/LCS/LCSD

Laboratory Control Sample Duplicate (LCSD)

Lab Sample ID: CI250513_1633.LCSD250512, Parent Sample ID: CI250513_1633.LCS250512

Run in Batch: CI250513_1633, Run Date: 05/13/2025 13:56, Prep Date: 05/12/2025, Matrix: SO, Dilution: 1

Analyte	Flags	% Rec	LCL	UCL	RPD	RPD CL
PFBA		105.2	70	140	2.3	30.0
PFMPA		101.7	30	140	1.4	30.0
FPrPA (3:3 FTCA)		103.4	45	130	4.5	30.0
PFPeA		112.3	60	150	0.5	30.0
PFMBA		110.7	60	150	4.5	30.0
4:2 FTSA		105.1	60	150	0.2	30.0
NFDHA		88.0	60	155	16.8	30.0
PFHxA		105.7	65	140	1.6	30.0
PFBS		105.8	65	145	1.6	30.0
HFPO-DA		102.5	70	145	2.4	30.0
FPePA (5:3 FTCA)		90.0	60	130	16.5	30.0
PFEESA		110.6	70	140	0.5	30.0
PFHpA		111.8	65	145	2.0	30.0
ADONA		109.4	70	160	4.3	30.0
PFPeS		105.2	55	160	3.5	30.0
6:2 FTSA		104.8	55	200	6.9	30.0
PFOA		97.8	70	150	3.5	30.0
PFHxS		104.4	60	150	1.9	30.0
FHpPA (7:3 FTCA)		107.8	60	150	6.6	30.0
PFNA		95.7	70	155	2.5	30.0
8:2 FTSA		99.8	70	150	1.9	30.0
PFHpS		101.4	65	155	3.3	30.0
PFDA		104.4	70	155	0.8	30.0
N-MeFOSAA		105.7	65	155	5.7	30.0
EtFOSAA		105.5	65	165	1.5	30.0
PFOS		101.3	65	160	4.9	30.0
PFUnDA		108.1	70	155	2.4	30.0
9CL-PF3ONS		109.1	70	150	3.8	30.0
PFNS		103.1	55	140	0.1	30.0
PFDODA		98.9	70	150	1.6	30.0
PFDS		100.7	40	155	4.5	30.0
PFTTrDA		93.8	65	150	3.6	30.0
11CL-PF3OUdS		99.7	45	160	2.5	30.0
FOSA		108.4	70	140	0.8	30.0
PFTeDA		101.3	65	150	3.3	30.0
PFDOS		86.6	25	160	12.6	30.0
NMeFOSE		100.9	70	140	3.8	30.0
NMeFOSAM		104.1	70	155	9.7	30.0
NEtFOSE		105.2	65	165	0.1	30.0
NEtFOSAM		99.1	70	140	2.3	30.0

REPORT TO

CHAIN OF CUSTODY RECORD

INVOICE TO

CONTACT NAME CINDY TOMASZEWSKI			
COMPANY City of Cadillac - WWTP			
ADDRESS 1121 Plett Rd			
CITY Cadillac		STATE MI	ZIP CODE 49601
PHONE NO. 231-775-2368		CELL NO.	
E-MAIL ADDRESS lab@cadillac-mi.net		P.O. NO.	
		QUOTE NO.	

CONTACT NAME		CINDY TOMASZEWSKI		X SAME	
COMPANY					
ADDRESS					
CITY				STATE	ZIP CODE
PHONE NO.		E-MAIL ADDRESS			

PROJECT NO./NAME QUARTERLY BIOSOLIDS	SAMPLER(S) - PLEASE PRINT/SIGN NAME
TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS ☒ STANDARD ☐ OTHER _____	
DELIVERABLES REQUIRED ☒ LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER _____	

MATRIX	W=WATER	GW=GROUNDWATER	WW=WASTEWATER	S=SOIL	L=LIQUID	SD=SOLID
CODE:	SL=SLUDGE	DW=DRINKING WATER	O=OIL	WP=WIPE	A=AIR	WS=WASTE

[illegible]

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)									
# Containers & Preservatives									
NONE	HCl	HNO ₃	H ₂ SO ₄	NaOH	MeOH	OTHER			
X							X	PFAS - MHD. 1/6/33	
							X	Total Solids	
							Certifications ☐ OHIO VAP ☐ Drinking Water ☐ DoD ☒ NPDES Project Locations ☐ Detroit ☐ New York ☒ Other <u>MICHIGAN</u> Special Instructions <u>EGLE Param. Lis</u> <u>3 RL's</u>		

RELINQUISHED BY: SIGNATURE/ORGANIZATION	<i>T. Loggins / City of Cadillac</i>	DATE	TIME
RECEIVED BY: SIGNATURE/ORGANIZATION		4-28-25	12
RELINQUISHED BY: SIGNATURE/ORGANIZATION		DATE	TIME
RECEIVED BY: SIGNATURE/ORGANIZATION		DATE	TIME

RELINQUISHED BY:		UPS		DATE		TIME	
SIGNATURE/ORGANIZATION				4-27-05		1000	
RECEIVED BY:		J. Helms		DATE		TIME	
SIGNATURE/ORGANIZATION				4-27-05		1000	
SEAL NO.	SEAL INTACT YES ☐ NO ☐	INITIALS	NOTES:	TEMP. ON ARRIVAL			
SEAL NO.	SEAL INTACT YES ☐ NO ☐	INITIALS	4.4°C				